

<page_start no = 1>

<sol_start id=1>

1. (1) Tonoplast

<sol_end>

<sol_start id=2>

2. (4) Soluble starch

<sol_end>

<sol_start id=3>

3. (1) Haploid set of chromosomes

<sol_end>

<sol_start id=4>

4. (2) H₂O

<sol_end>

<sol_start id=5>

5. (3) K⁺

<sol_end>

<sol_start id=6>

6. (4) Hydathodes

<sol_end>

<sol_start id=7>

7. (1) Telophase

<sol_end>

<sol_start id=8>

8. (3) GS and gs

<sol_end>

<sol_start id=9>

9. (3) Overall, water molecules moves from left to right.

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<sol_start id=10>

10. (4) Hot, dry and windy

<sol_end>

<sol_start id=11>

11. (3) stuoma

<sol_end>

<sol_start id=12>

12. Asters (2)

<sol_end>

<sol_start id=13>

13. (2) sex chromosomes

<sol_end>

<sol_start id=14>

14. (4) 25% TT, 50% tT, 25% tt

<sol_end>

<sol_start id=15>

15. (4) stop growing and die gradually of starvation

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<sol_start id=16>

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16. Mendel's first law of inheritance are :-

⇒ Law of dominance : The first law of inheritance is law of dominance means which suppressed the receive then get dominant over it and expressed own character.

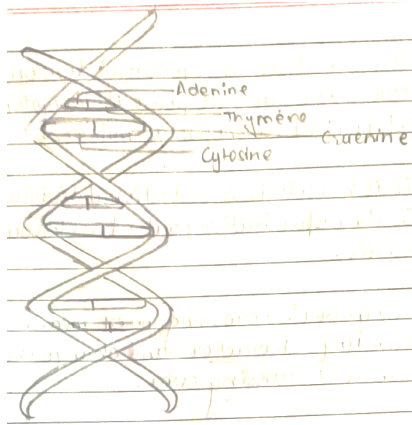
for example Tall plant (TT) cross over short plant (tt)



In f_1 phase all plants are tall.

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(ii)

The DNA (Double helical model) of four nitrogenous product adenine, Thymine, Guanine and Cytosine.

Whereas Adenine made double hydrogen bond with Thymine (T) then, Guanine made triple hydrogen bond with cytosine.

$A = T$

$G \equiv C$

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(iii) (a) The meaning of destarched plants is plant leaves without starch. It can be destarched by the iodine solution.

(b) (i) Take a plant's leaves without the presence of starch.

(ii) Then, cover a half part of the leaf with the help of a black strip.

(iii) Leave the plants in a light place.

(iv) Then, use iodine solution where starch is not present; it converted into Brown in colour.

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(iv) (a) Mitosis : Mitosis is equational division and it completes these stages in two phases (i) karyokinesis (ii) cytokinesis. It is known as mitosis.

(b) Meiosis : Meiosis is reductional division and present in the reproductive part. It is known as meiosis.

- (c) Crossing over : crossing over means crossing a contrasting character and change the genotypes is known as crossing over.
- (d) Karyokinesis : Karyokinesis is division of nucleus and present in mitosis.
- (e) Interphase : Interphase are resting phase of mitosis and involve three steps G₁, S and G₂ this known as interphase.

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17. (i) open stomata

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(ii) A-epidermis B-Guard cell C- stomata O-nucleus

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(iii) During transpiration stomata will open and give a coating effect the surface area of leaf on other hand during photosynthesis stomata will closed.

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(iv) The function of stomata are followed :

⇒ It regulates the exchanges of gases (CO₂ and O₂)

⇒ It maintain the concentration level of water.

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(v) Three kind of transpiration are :

⇒ Stomatal transpiration :- Transpiration through minute opening of stomata in the leave is known as stomatal transpiration.

⇒ Cuticles transpiration : Transpiration through cuticles present on the leaves is known as cuticular transpiration.

⇒ Lenticular transpiration :- Transpiration through old barle tree present at the stem is known as lenticular transpiration.

* Minimum transpiration – lenticular

* Maximum transpiration – stomatal

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18. (i) Prophase

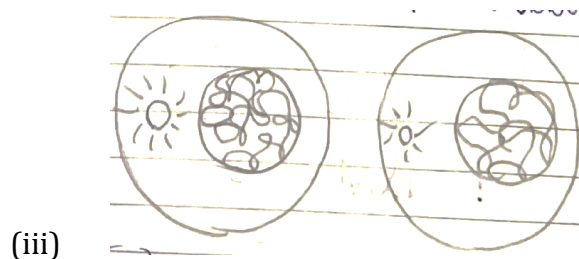
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(ii) Centromere, (2) chromatids, (3) spindle fibre

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nuclear membrane appear

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(iv) (a) mitosis division

(b) meiosis division

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(v)

Mitosis	Meiosis
* 2 daughter cell produced	* 4 daughter cell produced
* equational division	* reductional division
* It present somatic and vegetative part	* It present on reproductive part.
* It is diploid	* It is haploid

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<sol_start id=21>

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21. (i) A – hypotonic solution

B – hypertonic solution

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(ii) A – deplasmolysis

B – plasmolysis

<sub_id end>

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(iii) hypotonic (cell A)

Hypertonic (cell B)

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(iv) Osmotic pressure, root pressure

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(v) The path of ascent of sap are :

Cohension : water molecules attached with water molecules.

Adhesion : water molecules attached with other substance.

Root pressure : The pressure exerted by the root for the movement of water molecules.

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